

Network Cable Types and ToolsTheory

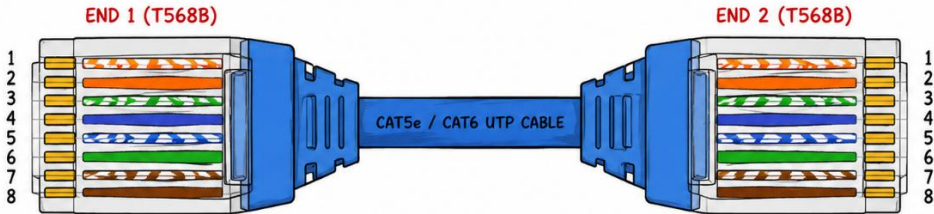
- 1) List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Difference	Ethernet Cable	Fiber Optic Cable
1. Data Transmission	Uses electrical signals through copper wires to transfer data.	Uses light signals through glass/plastic fibers to transfer data.
2. Speed & Distance	Suitable for shorter distances (usually up to 100 meters) with good speed.	Supports much higher speeds and can cover very long distances (kilometers).
3. Cost & Installation	Cheaper, easier to install and maintain.	More expensive and requires specialized equipment for installation.

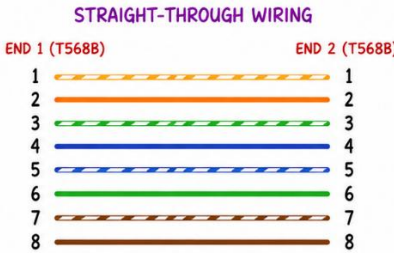
- 2) Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

STRAIGHT-THROUGH ETHERNET CABLE (T568B WIRING STANDARD)

Both ends use T568B standard (Same pinout on both ends)



T568B WIRING (Both Ends)		
PIN NO.	WIRE COLOR	FUNCTION
1	White / Orange	Transmit Data +
2	Orange	Transmit Data -
3	White / Green	Receive Data +
4	Blue	Unused
5	White / Blue	Unused
6	Green	Receive Data -
7	White / Brown	Unused
8	Brown	Unused



NOTE:

- Straight-through cables are used to connect different types of devices. (e.g., PC to Switch, Router to Switch)

Orange Pair	Blue Pair
Green Pair	Brown Pair

- 3) Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Steps to Attach an RJ45 Connector to a CAT6 Cable:

- i. **Prepare the tools:**
Collect a CAT6 Ethernet cable, RJ45 connectors, cable stripper, wire cutter, and RJ45 crimping tool.
- ii. **Cut and strip the cable:**
Cut the CAT6 cable to the required length and remove about 2–3 cm of the outer protective jacket using a cable stripper.
- iii. **Separate the wires:**
Untwist the four twisted wire pairs and arrange the 8 small wires according to the **T568B color standard:**
 - White/Orange
 - Orange
 - White/Green
 - Blue
 - White/Blue
 - Green
 - White/Brown
 - Brown
- iv. **Straighten and trim the wires:**
Align all wires in the correct order and cut them evenly so they can fit properly inside the RJ45 connector.
- v. **Insert wires into RJ45 connector:**
Carefully insert the wires into the RJ45 plug, making sure each wire reaches the correct pin and the cable jacket goes inside the connector.
- vi. **Crimp the connector:**
Place the RJ45 connector into the crimping tool and press firmly to lock the pins onto the wires.
- vii. **Test the cable:**
Use an Ethernet cable tester to check if all pins are connected correctly and the cable works properly.

4) Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

- For a gaming café with 10 PCs, I would use Ethernet cables (CAT6) because they provide high speed, low latency, and are affordable for short distances inside the café. I would use an RJ45 crimper to attach RJ45 connectors to the Ethernet cables. Fiber optic cables are faster but more expensive and unnecessary for a small LAN gaming setup. Ethernet cables are easier to install, maintain, and provide enough performance for online gaming and LAN matches.